

Designing Gamified-Service Towards User Engagement and Service Quality Improvement

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Abstract--- Organizations find new challenge which is related to employees' engagement along service desk implementation. Gamified-service is defined as service desk which integrated Information Technology Infrastructure Library (ITIL) and gamification model. This integration enables fun, challenge, and reward to boost employees' motivation. Pressure in the workplace and employees' personal issues become the main trigger to decrease productivity. According to this reason, we need to add persuasive approach along implementation of ITIL best practices. This research focused on designing gamified-service prototype in order to help organization build game-like workplace environment. We believe that gamified-service will help employees do their job with fun way.

Keyword--- gamification; ITIL; service desk; user engagement; web-based application

I. INTRODUCTION

Nowadays there are a lot of organizations use information technology (IT) to support their business. It makes the organizations spend big amount of investment for IT. But, even they pay expensive cost for IT, it can help them to avoid them from incident of IT. Incident IT is able to slow the business operation. When business operation is corrupted, there will be delayed activities. It will cause negative impacts for organizations. According to this matters, organizations which operate and demand on IT need a function that enables communication line to report incident IT and give solution.

Service desk is well known function which acts as first communication point between user and IT department. It handles incident, request, event, and problem of IT. This function will be required to be implemented at every organization which runs IT as tool to help their business [1].

Along the implementation of service desk, organization need to follow best practices. Information Technology Infrastructure Library (ITIL) is the most popular best practices to implement good service desk as ITSM feature. According to ITIL, service desk runs processes. These processes are important because they are directly connected to customers' needs.

Along the implementation of ITIL, there is new challenge which is related to engagement of service desk staffs. According to previous research, there is positive correlation between user engagement and successful IT implementation [2]. It is supported by other research which says that effective ITIL implementation is related to user engagement [3]. Therefore, we need other approach to boost service desk staffs' motivation.

In this research, we use gamification model as persuasive approach. We believe that gamification model will help service desk to do their task with fun way and ITIL as best practices will ensure that service desk staffs follow well documented and standard procedures.

II. LITERATURE REVIEW

A. ITIL

ITIL is document that presents best practices to manage the implementation of IT service. ITIL explains the detail of management and IT operations such as incident management, problem management, change management, configuration management, and availability management. The latest version of ITIL is ITIL 3.0 which has been published in 2007 by Office of Government (OGC). It contains 5 phases of service lifecycle.

- 1) Service strategy
- 2) Service design
- 3) Service transition
- 4) Service operation
- 5) Continual service improvement

B. Service desk

To run ITIL processes, we need a function. Service desk is a function to provide communication between users and IT employees. Service desk roles and responsibilities are senior service desk manager, service desk manager, service desk supervisor, and service desk analyst [4]. Service desk primary tasks are providing solution of IT infrastructure incident, request, problem, etc. Moreover, service desk is a single and first line which communicate with users in order to information about status of incident, request, problem, etc.

C. Incident management

Process is defined as a set of structured activities and tasks that produce a specific goal [5]. ITIL processes have been specified. Such as incident management, it has been specialized to manage incident. Incident management is often the first process when introducing the ITIL framework to a service desk, and offers the most immediate and highly cost reduction and quality gains [6]. Below are activities which are operated by incident management [7][8]:

- 1) Incident identification
- 2) Incident logging
- 3) Categorization incident
- 4) Prioritization incident
- 5) Diagnostic
- 6) Escalation
- 7) Further diagnostic
- 8) Resolution
- 9) Close incident

D. Gamification

Many organizations have implemented gamification model in order to motivate their employees while doing their tasks. Last four years, gamification has been discussed. In 2011, for the first time gamification is defined by Deterding, it is defined as the used of game design elements in non-game context [9].

Gamification is triggered by research about human's brain by scientists around the world. They agree that challenge – achievement - reward loop promotes production of dopamine. It can create satisfaction and reinforce desire to play [10].

Many applications have been using gamification, like Nike+ Running, GoJek, and Foursquare. Gamification helps the organization to gain customer 's satisfaction. It is not a user interface design, it is about how to reach the goal with fun way. Gamification enables win – win solution between organization and employee [11]. Organization can reach their goal, and employees are able to get the internal satisfaction, like fully engaged system.

Based on research, job satisfaction, high loyalty, and productivity of employees are stimulated by good internal service and organization policy [12]. It means issues which are related to employees satisfaction is critical in helping organizations reach their goals and business such as satisfying their customers.

E. User Engagement Measurement

User engagement is defined as the quality of the user experience that emphasizes the positive aspects of the interaction. It is not just about usability, but it also contains how users invest time, attention, and emotion when they are connected to the system [11]. We can identify each attributes of user engagement degree by using user engagement measurement. According to the

previous research, user engagement attributes are focused attention, aesthetic, novelty, perceived usability, durability, felt involvement[13].

III. DESIGNING GAMIFIED-SERVICE

Gamified- service is web-based service desk system that gamifying the processes which are mentioned in ITIL service operation documents. In this paper, we will introduce the prototype of gamified – service which runs function service desk and process incident management. Based on ITIL, there are 3 main roles of service desk, service desk analyst (SDA), service desk supervisor (SDS), service desk manager (SDM).

First, SDA acts as the first level of service desk. It responsables to provide quick solution for reported incidents, entry all details about incident, do basic analyze, and deliver solution from upper level to customer if the incident report is escalated.

Second, SDS as second level who ensures the skill levels of staffs are maintained and representing SDAs at meetings, SDS becomes the escalation point when SDA finds difficulties or controversial calls.

Third, SDM as escalation point from SDS level. SDM provides solution for incidents which can not be handled by first and second level. Moreover, he responsables to do advance analyze of business impact which caused by incidents and report it to the board. Fig 1 shows the use case diagram for service desk roles.

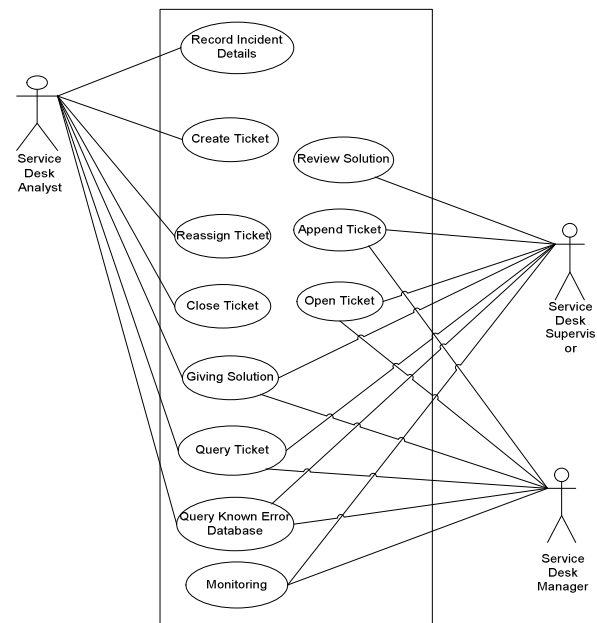


Fig.1 Use Case Diagram of Service Desk Roles

There are seven main features of gamified- service prototype, namely, (1) record customer, (2) record incident, (3) solution database (4) known error database (KEDB), (5) escalation, (6) close incident, (7) leaderboard.

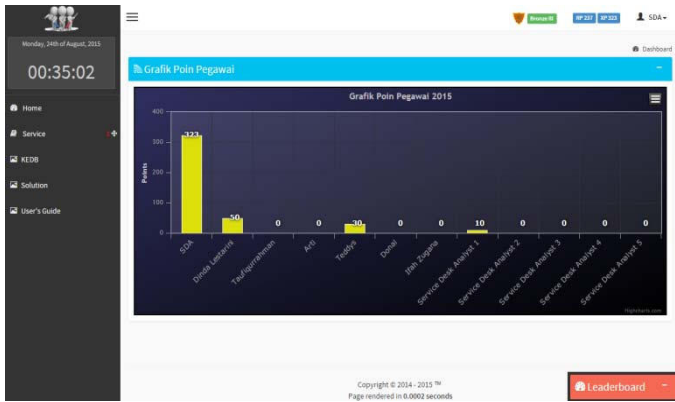


Fig.2. Gamified- Service Home Page

First, customer will report the incident to SDA and he will create a new data customer if customer does not exist in database.

Second, when the record of customer is already saved, SDA entries the details of incident such as category of incident, description of incident, time and date of incident, level of urgency, etc.

Third, SDA checks solution database to find solution. If expected solution can not be found in solution database, he will provide solution based on his skills and knowledge. This activity is also the mechanism of input solution. When solution of incident is first time recorded to database, it is not directly inserted to solution database. It needs supervisor to review the solution. Once it is accepted, it will be recorded as solution.

Fourth, incidents which are reported more than once will be recorded on KEDB. This features will help organization to find the weakness of IT Service and enable the opportunity to evaluation and improvement. In this feature, we will see incident records, their solutions, and how many times they reported.

Fifth, SDA can use solution database to deliver the solution and his own skills and knowledge, if customer is satisfied, SDA is able to close the incident. In contrary, if SDA finds the customer doubts, it means SDA has to provide more information and find suitable solution. In order to get support and help, SDA can run escalation feature. This feature enables SDA to create incident ticket and forward it to upper level, SDS and SDM. SDS and SDM are responsible for opening the escalated ticket. Previous research finds that the key success of ITIL implementation is user engagement with good support from upper level [3]. It means SDS and SDM are able to work together with SDA and direct them if they find problem to solve incident. If SDS can not fix incident, SDS escalated incident ticket to SDM. If SDM finds difficulties and he needs more time to fix they will forward the incident ticket to problem management, which means the incident ticket is failed. Moreover, incident ticket will fail if the incident ticket is not open, and it causes SDA loses RP. But, the gamified-service does not include the punishment for irresponsible SDS and SDM. This matters will be charged to upper level of organization.

Sixth, when solution for incident is delivered, SDA close the incident management process. If the incident management is closed before time ends, SDA will get points. and in contrary if it is closed after time has ended, SDA will lose points. In this system, points are categorized by two types, redeemable points, and experience point (XP). Service is failed if the incident ticket is not opened and solution is not delivered to customer until the time ends. Table 1 shows description about urgency level and estimated time to solve the incident.

TABLE 1 URGENCY LEVEL AND TIME TO SOLVE

Urgency level	Estimated time to solve incident
1 = Low	12 – 23 hours 59 minutes 59 seconds
2 = Middle	6 – 11 hours 59 minutes 59 seconds
3 = High	0 – 5 hours 59 minutes 59 seconds

When the time ends, incident ticket is failed and SDA will lose promised RP. In contrary, even the incident ticket is failed, XP keeps gaining because even incident ticket is failed, SDA gets experience. In other case, when SDA delivers the solution to customer on scheduled time, RP and XP will gain. RP can be exchanged to desirable contents, such as bonus, shopping voucher, etc. While XP represents SDA real experience. Table 1 shows the XP rules and table 2 shows RP rules.

TABLE 2 RULES FOR XP

Condition	Urgency	XP gains when incident ticket is success	XP gains when incident ticket is failed
XP current = 0-750	1	+10	+1
	2	+12	
	3	+20	
XP current = >751	1	+5	
	2	+10	
	3	+15	

TABLE 2 RULES FOR RP

Condition	Urgency	RP gains when incident ticket is success	RP gains when incident ticket is failed
RP current = 0-500	1	+10	-5
	2	+12	-6
	3	+20	-10
RP current	1	+5	-3

= >751	2	+10	-5
	3	+15	-8

Seventh, leaderboard shows SDAs with highest XP and RP. It helps organization to create competition environment in workplace. There are many game elements that can be used, in this prototype we use points as core element. Points are able to be converted to other game elements, when SDA has points, then organization can add other elements such as avatar, badges, exclusive contents, virtual contents, etc. These elements depend on points, system needs points to upgrade badges, so does when system needs to give exclusive contents to SDA.

Along the implementation of gamified-service, we need to identify what are indicators of success. It also helps organizations to get information about how far they have gone. Tabel 4 shows the basic evaluation indicators for service desk [13]. In order to find if users are engaged with gamified-service, we need to use User Engagement Scale (UES) [12].

We suggest a gap analysis between pre-implementation of gamified-service and post implementation. After do a gap analysis, organizations are able to discover if the gamified-service reaches the target. Additionally, gap analysis enables organization to find new challenge in IT service improvement.

TABLE 4 BASIC EVALUATION INDICATORS FOR SERVICE DESK

Indicators	Explanation
Response time	Response user's calls as soon as possible
Call rate	Rate of request for help (RFH) that comes from service desk
Records of request call	Numbers of the events and incidents recorded by service desk
Rate of complaints	Assessing the service desk whether handling incident
Accuracy of incident classification	Assessing the service desk whether handling accidents by the defined categories
Growth rate of desk incident	The rate that incidents could not be processed
Rate of incident handling	Remote processing rate when service desk receives the RFH
First call completion rate	Connected rate when user first calls

IV CONCLUSION

Service desk employees must be fully engaged to the system because they responsible to communicate with users in order to help users fix incidents, problem, etc. If they can not engage to the system, then it will cause slow response to the request, event-handling, incident, etc. Engagement deals with employees' motivation, involvement, and participation. According to this matter, we propose a design of gamified-service which integrate ITIL best practices and gamification model. By using gamification in service desk, the engagement

will be increased and may return benefits to organization such as internal satisfaction, customer trust, etc. This design can be used to do some future research in gamified- service, like implementing mobile gamified-service, gamification system for customer, and finding what are the most influence user engagement attributes for gamified service.

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